

OPENHMI REFERENCE MODULE

TC3568M

Rockchip RK3568

English Technical Reference Manual

A higher-performance option for premium embedded HMI, multi-interface display systems, edge applications and products that need PCIe, USB 3.0 or larger memory configurations.

Revision 1.1 | August 2026

Supplier-neutral reference compiled from the supplied module documentation. Confirm the exact hardware revision, pin mapping, BSP status and commercial configuration before design-in.

1. Positioning and selection

This manual summarizes the module from a project-selection perspective. It is intended to help product teams decide whether the platform fits their HMI architecture before requesting detailed carrier-board, software and supply-chain information.

When this module is a good fit

- Premium industrial HMI, digital signage and medical or commercial equipment
- Linux or Android products requiring Cortex-A55 processing and stronger graphics
- Designs requiring HDMI, eDP, MIPI DSI, LVDS or parallel RGB choices
- Systems needing PCIe, USB 3.0, dual Ethernet or larger memory/storage options

When to consider another path

- Simple control panels where MCU-class or lower-cost Linux hardware is sufficient
- Designs unable to support a 3.3 V high-current module power rail and careful layout
- Projects that do not benefit from the added interface and performance capability

Selection rule

Choose the module only after checking display interface, graphics load, boot-time target, operating-system baseline, carrier-board complexity, production lifecycle and the engineering capability available to your team.

2. Core specifications

Processor	Rockchip RK3568; quad-core Arm Cortex-A55
Graphics	Mali-G52; OpenGL ES 3.2, OpenCL 2.0 and Vulkan 1.1 class support
Memory	LPDDR4/LPDDR4X, typically 2 GB; documented configurations up to 8 GB
Storage	eMMC, typically 8 GB; documented support up to 128 GB
Operating systems	Linux and Android 11 support are documented
Module interface	192 castellated edge pins
Input power	3.2 V to 3.5 V DC; 3.3 V nominal; selected USB VBUS-related functions use 5 V

Configuration note

Memory, storage, wireless and some peripheral options are configuration-dependent. The values above describe documented capability, not a guaranteed stock configuration. Confirm the required combination before quotation or schematic release.

Software baseline

Operating-system support depends on the available BSP, kernel branch, display driver, touch controller, GPU stack and peripheral drivers. Validate the actual software package on target hardware before committing the product schedule.

3. Display, video and touch

Panel outputs	Parallel RGB, LVDS, MIPI DSI and eDP; functions share multiplexed resources
External video	HDMI 2.0
Resolution	Module documentation lists display paths up to 2560 x 1600; validate the exact interface mode
Multi-display	Two independent output paths are documented for selected interface combinations
Camera/video input	MIPI CSI and up to 16-bit parallel DVP; BT.656/BT.1120 support is documented
Touch	I2C interface suitable for common capacitive-touch controllers

Display integration checks

- Confirm that the required panel timing, lane count and voltage level are supported by the selected pin-multiplexing mode.
- Check whether dual-display use reduces access to camera, Ethernet, storage or general-purpose I/O functions.
- Validate backlight power, enable and PWM control on the carrier board; these are not supplied by the display data interface alone.
- Confirm the touch-controller interface, reset/interrupt pins, firmware and display bonding stack for the intended panel.
- Run the real UI workload on the target resolution to verify frame rate, memory bandwidth and thermal behavior.

4. Connectivity and I/O

USB	1 x USB 3.0 OTG, 1 x USB 3.0 host and 2 x USB 2.0 host
Expansion	1 x PCIe 2.0 and 1 x PCIe 3.0 interface; SATA-related capability is SoC-dependent
Ethernet	One 10/100/1000 MAC and one 10/100 MAC; dual-port operation is documented
Serial buses	Up to 9 UART, 2 SPI and 4 TWI/I2C through multiplexed pins
Control	Up to 6 PWM, ADC inputs, infrared controller and RTC support
Audio	Microphone input, stereo output, I2S and S/PDIF options
Wireless	Optional Wi-Fi/Bluetooth; cellular and GNSS can be added through carrier-board expansion

Pin-multiplexing warning

Interface counts describe the documented processor and module capability. Several functions share pins, clocks or high-speed resources and therefore cannot always be used together. Build a project-specific I/O allocation table before starting the carrier-board schematic.

Request before layout

Obtain the current pin table, carrier-board reference schematic, PCB footprint, power sequence and BSP release note for the exact module revision selected for the project.

5. Mechanical, power and design-in

Module size	50 mm x 50 mm
Overall height	Approximately 3.2 mm
PCB thickness	1.2 mm
Carrier keep-out	A carrier-board cutout is required under the bottom-side component area

Power and handling

- Do not hot-plug the module or connect/disconnect it while the carrier board is powered.
- Use a regulated supply with startup margin for display, USB, Ethernet, wireless and storage peak current.
- Follow the documented power sequence, reset timing and signal-voltage domains.
- Provide airflow or heat spreading after measuring the real application workload and enclosure temperature.
- Protect the assembly from condensation, conductive contamination, ESD and mechanical stress.

Project validation checklist

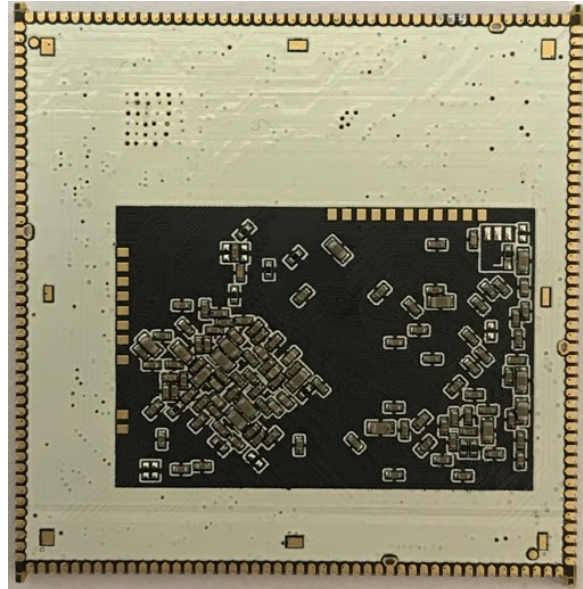
- Exact module revision and production configuration confirmed
- Display and touch panel tested with the intended cable and bonding stack
- All pin conflicts resolved in a carrier-board I/O matrix
- BSP, boot media, update method and recovery path verified
- Thermal, EMC, ESD, power interruption and long-duration operation tested
- Lifecycle, lead time and alternative-component strategy reviewed

6. Hardware views and reference drawings

The photographs below are reproduced from the supplied module documentation. Supplier marks have been redacted while component placement, connectors and edge contacts are left unchanged. Photographs are for identification only and must not be scaled for PCB or enclosure design.



Module top view



Module bottom view

System application block diagram

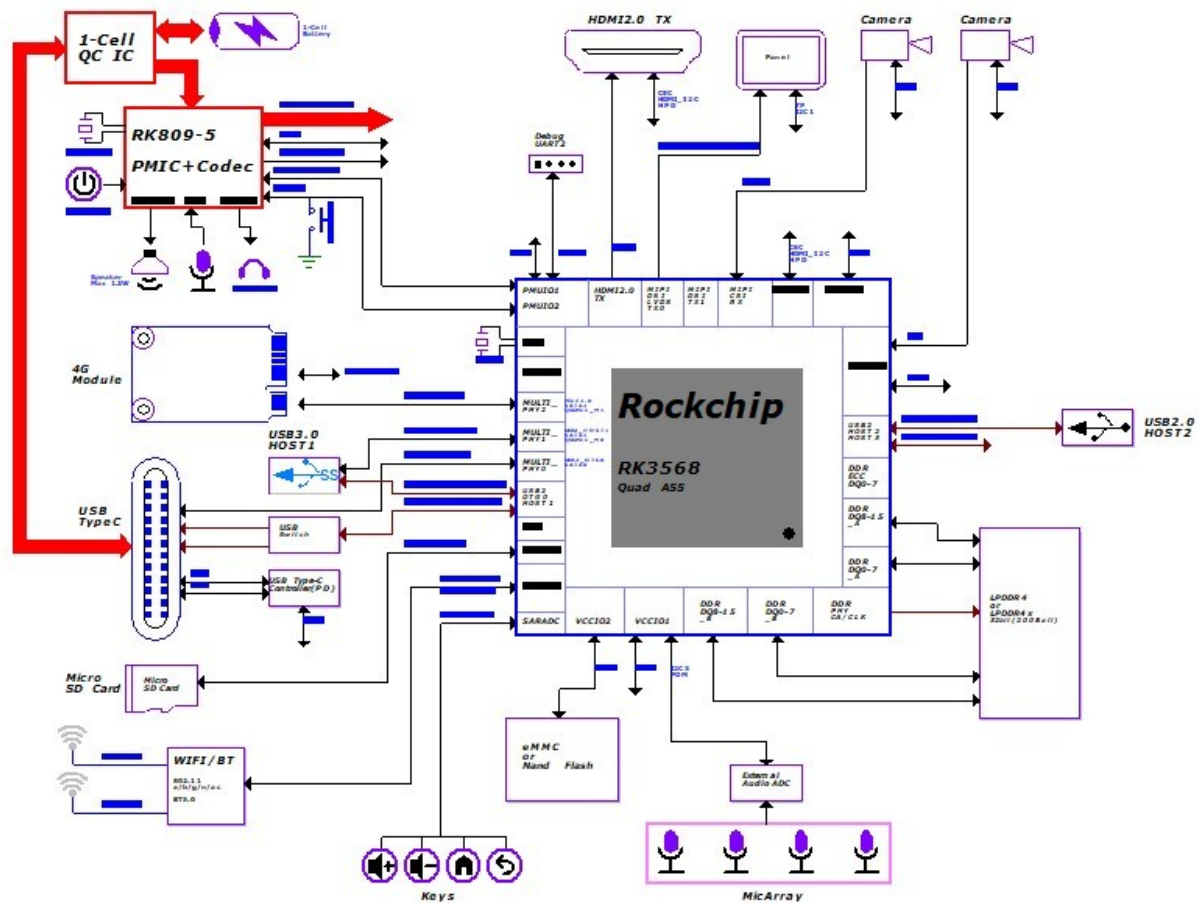


Figure: System application block diagram

Module pin-side orientation map

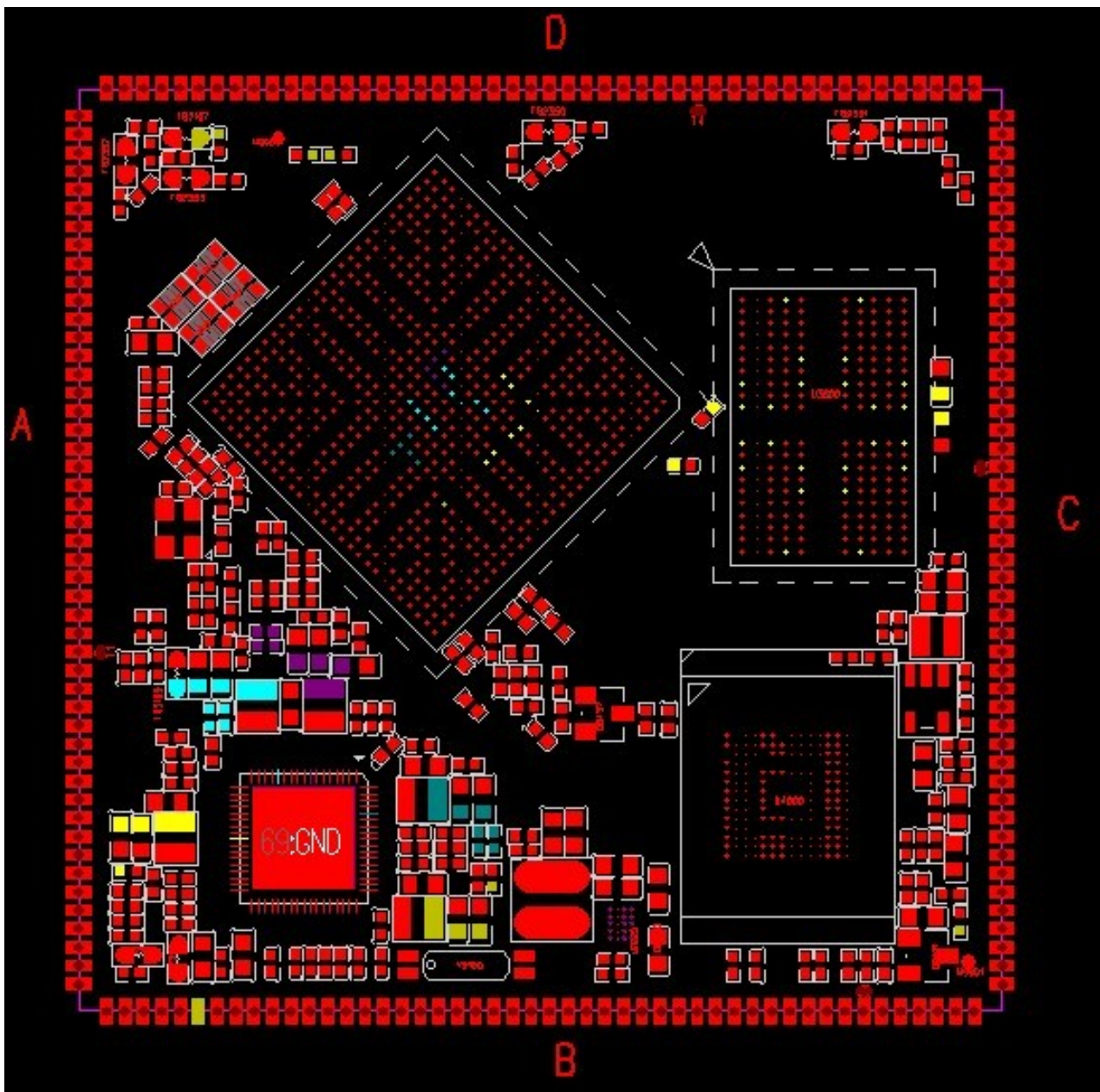


Figure: Module pin-side orientation map

7. Pin definitions and multiplexing reference

The following pages reproduce the original engineering pin tables at high resolution. Signal identifiers and alternate-function names are preserved exactly to reduce transcription risk. Use them as an evaluation reference; the controlled pin table for the selected hardware revision remains the design authority.

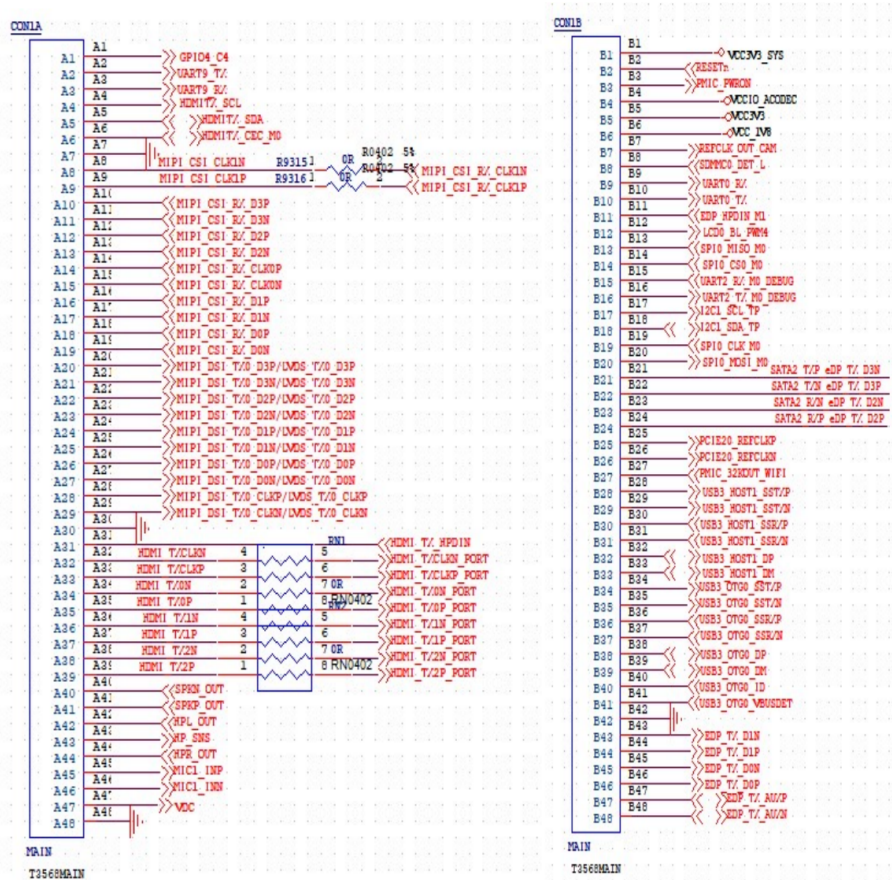
English legend for recurring source labels

PIN / GPIO	Physical module pin and associated GPIO bank or signal
Function 1...n	Selectable alternate function through SoC pin multiplexing
Input / Output / I/O	Signal direction: input, output, or bidirectional
Power input / Power output	Supply rail entering or leaving the module
GND	Power and signal ground
NC / Floating	Do not connect unless a controlled revision document states otherwise
Highlighted cells	Default or reserved module functions in the source table; confirm before reassignment

Important

Never route a carrier board from a web manual alone. Request the revision-matched pin spreadsheet, schematic, footprint and power-domain table before schematic sign-off.

Pin definition reference - sheet 1 of 11



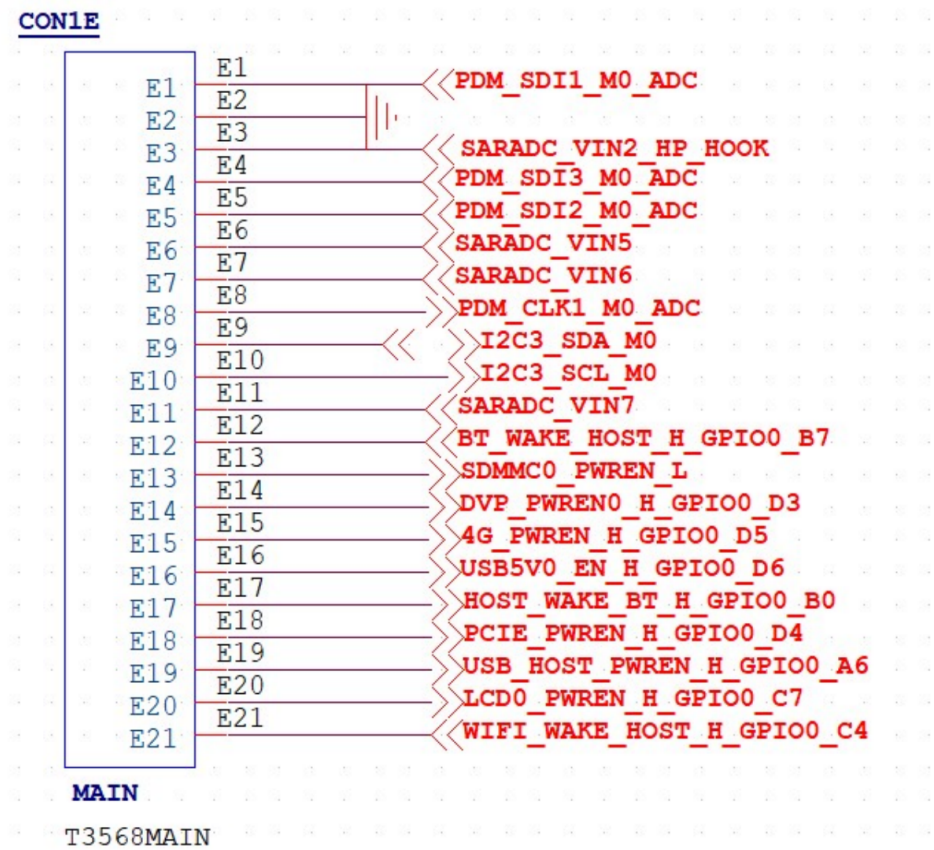
Original engineering table, source page 14. Signal names are preserved; see the English legend on the preceding page.

Pin definition reference - sheet 2 of 11

CON1C			CON1D		
C1	C1	>SERVIC0_D1	D1	D1	<GPAC1_MCLKINOUT_M1
C2	C2	>SERVIC0_D0	D2	D2	<GPAC1_R/CLK_M1
C3	C3	>SERVIC0_CLK	D3	D3	<GPAC1_R/DV_CRS_M
C4	C4	>MCC3V3_SD	D4	D4	<GPAC1_R/D0_M1
C5	C5	>SERVIC0_CMD	D5	D5	<GPAC1_R/D1_M1
C6	C6	>SERVIC0_D3/ARMUTAG_TMS	D6	D6	<GPAC1_R/D2_M1
C7	C7	>SERVIC0_D2/ARMUTAG_TCK	D7	D7	<GPAC1_R/D3_M1
C8	C8	>MMC_KEY	D8	D8	<GPAC1_T/CLK_M1
C9	C9	>ETH0_REFCLK0_25M	D9	D9	<GPAC1_T/EN_M1
C10	C10	>GPAC0_MCLKINOUT	D10	D10	<GPAC1_T/D0_M1
C11	C11	>GPAC0_R/CLK	D11	D11	<GPAC1_T/D1_M1
C12	C12	>GPAC0_R/DV_CRS	D12	D12	<GPAC1_T/D2_M1
C13	C13	>GPAC0_R/D0	D13	D13	<GPAC1_T/D3_M1
C14	C14	>GPAC0_R/D1	D14	D14	<GPAC1_MDIO_M1
C15	C15	>GPAC0_R/D2	D15	D15	<GPAC1_MDC_M1
C16	C16	>GPAC0_R/D3	D16	D16	<GPAC1_MDC_M1
C17	C17	>GPAC0_T/CLK	D17	D17	<GPAC1_INT
C18	C18	>GPAC0_T/EN	D18	D18	<GPAC1_RSTn
C19	C19	>GPAC0_T/D0	D19	D19	<GPAC0_INT
C20	C20	>GPAC0_T/D1	D20	D20	<GPAC0_RSTn
C21	C21	>GPAC0_T/D2	D21	D21	<PCIE30/2_FRSNT_L
C22	C22	>GPAC0_T/D3	D22	D22	<PCIE30/2_CLRREQn_M1
C23	C23	>GPAC0_MDIO	D23	D23	<PCIE30/2_WAKEn_M1
C24	C24	>GPAC0_MDC	D24	D24	<PCIE30/2_PERSTn_M1
C25	C25	>UART0_RTSn_M0	D25	D25	<GPIO3_A0
C26	C26	>UART0_CTSn_M0	D26	D26	<SPI1_CS0_M1
C27	C27	>UART0_T/_M0	D27	D27	<GPIO3_A2
C28	C28	>UART0_R/_M0	D28	D28	<I2S3_SCLK_M0
C29	C29	<BATT_TC_L	D29	D29	<I2S3_LRCK_M0
C30	C30	<SARADC_VINO_KEY/RECOVERY	D30	D30	<I2S3_SDO_M0
C31	C31	<USB2_HOST2_DP	D31	D31	<I2S3_SDI_M0
C32	C32	<USB2_HOST2_DM	D32	D32	<PCIECLKIC_OE_H
C33	C33	<USB2_HOST3_DP	D33	D33	<ETH1_REFCLK0_25M_M0
C34	C34	<USB2_HOST3_DM	D34	D34	<UART4_R/_M1
C35	C35	<SERVIC2_D1_M0	D35	D35	<UART4_T/_M1
C36	C36	<SERVIC2_D0_M0	D36	D36	<I2CS_SCL_M0
C37	C37	<SERVIC2_CLK_M0	D37	D37	<I2CS_SDA_M0
C38	C38	<SERVIC2_CMD_M0	D38	D38	<GPIO3_B5
C39	C39	<SERVIC2_D3_M0	D39	D39	<GPIO3_B6
C40	C40	<SERVIC2_D2_M0	D40	D40	<UART3_T/_M1
C41	C41	<SERVIC2_D1_M0	D41	D41	<UART3_R/_M1
C42	C42	<WIFI_WAKE_HOST	D42	D42	<SPI1_NDSIF_M1
C43	C43	<WIFI_REG_ON	D43	D43	<SPI1_MISO_M1
C44	C44	<CIF_CLKOUT	D44	D44	<SPI1_CLK_M1
C45	C45	<CAN2_T/_M0	D45	D45	<UART7_T/_M1
C46	C46	<CAN2_R/_M0	D46	D46	<UART7_R/_M1
C47	C47	<I2C4_SCL_M0	D47	D47	<CAN1_RX_M1
C48	C48	<I2C4_SDA_M0	D48	D48	<CAN1_TX_M1

Original engineering table, source page 15. Signal names are preserved; see the English legend on the preceding page.

Pin definition reference - sheet 3 of 11



2.4 PIN 功能应用说明

- 2.4.1 标注为默认软件 SDK 调试好的功能应用。
- 2.4.2 据需求选择相应的功能应用。
- 2.4.3 E 端口在核芯板的底部。

A 端口如下:

PIN	GPIO 功能	IO 电平	IO 状态	IO 多功能复用定义
A1	GPIO4_C4	IO=3V3	输入输出	EDP_HPDIN_M0/SPDIF_TX_M2/SATA2_ACT_LED/PCIE 30X2_PERSTn_M2/I2S3_LRCK_M1/GPIO4_C4_d
A2	UART9_TX	IO=3V3	输入输出	PWM12_M1/SPI3_MISO_M1/SATA1_ACT_LED/UART9

Original engineering table, source page 16. Signal names are preserved; see the English legend on the preceding page.

Pin definition reference - sheet 4 of 11

				_TX_M1/I2S3_SDO_M1/GPIO4_C5_d
A3	UART9_RX	IO=3V3	输入输出	PWM13_M1/SPI3_CS0_M1/SATA0_ACT_LED/UART9_RX_M1/I2S3_SDI_M1/GPIO4_C6_d
A4	HDMITX_SCL	IO=3V3	输入输出	HDMITX_SCL/I2C5_SCL_M1/GPIO4_C7_u
A5	HDMITX_SDA	IO=3V3	输入输出	HDMITX_SDA/I2C5_SDA_M1/GPIO4_D0_u
A6	HDMITX_CEC_M0	IO=3V3	输入输出	HDMITX_CEC_M0/SPI3_CS1_M1/GPIO4_D1_u
A7	GND	GND		
A8	MIPI_CSI_RX_CLK1N		输入	MIPI_CSI_RX_CLK1N
A9	MIPI_CSI_RX_CLK1P		输入	MIPI_CSI_RX_CLK1P
A10	MIPI_CSI_RX_D3P		输入	MIPI_CSI_RX_D3P
A11	MIPI_CSI_RX_D3N		输入	MIPI_CSI_RX_D3N
A12	MIPI_CSI_RX_D2P		输入	MIPI_CSI_RX_D2P
A13	MIPI_CSI_RX_D2N		输入	MIPI_CSI_RX_D2N
A14	MIPI_CSI_RX_CLK0P		输入	MIPI_CSI_RX_CLK0P
A15	MIPI_CSI_RX_CLK0N		输入	MIPI_CSI_RX_CLK0N
A16	MIPI_CSI_RX_D1P		输入	MIPI_CSI_RX_D1P
A17	MIPI_CSI_RX_D1N		输入	MIPI_CSI_RX_D1N
A18	MIPI_CSI_RX_D0P		输入	MIPI_CSI_RX_D0P
A19	MIPI_CSI_RX_D0N		输入	MIPI_CSI_RX_D0N
A20	MIPI_DSI_TX0_D3P		输出	MIPI_DSI_TX0_D3P/LVDS_TX0_D3P
A21	MIPI_DSI_TX0_D3N		输出	MIPI_DSI_TX0_D3N/LVDS_TX0_D3N
A22	MIPI_DSI_TX0_D2P		输出	MIPI_DSI_TX0_D2P/LVDS_TX0_D2P
A23	MIPI_DSI_TX0_D2N		输出	MIPI_DSI_TX0_D2N/LVDS_TX0_D2N
A24	MIPI_DSI_TX0_D1P		输出	MIPI_DSI_TX0_D1P/LVDS_TX0_D1P
A25	MIPI_DSI_TX0_D1N		输出	MIPI_DSI_TX0_D1N/LVDS_TX0_D1N
A26	MIPI_DSI_TX0_D0P		输出	MIPI_DSI_TX0_D0P/LVDS_TX0_D0P
A27	MIPI_DSI_TX0		输出	MIPI_DSI_TX0_D0N/LVDS_TX0_D0N

Original engineering table, source page 17. Signal names are preserved; see the English legend on the preceding page.

Pin definition reference - sheet 5 of 11

	_DON			
A28	MIPI_DSI_TX0_CLKP		输出	MIPI_DSI_TX0_CLKP/LVDS_TX0_CLKP
A29	MIPI_DSI_TX0_CLKN		输出	MIPI_DSI_TX0_CLKN/LVDS_TX0_CLKN
A30	GND	GND		
A31	HDMI_TX_HP_DIN		输入	HDMI_TX_HP_DIN
A32	HDMI_TXCLKN_PORT		输出	HDMI_TXCLKN_PORT
A33	HDMI_TXCLKP_PORT		输出	HDMI_TXCLKP_PORT
A34	HDMI_TX0N_PORT		输出	HDMI_TX0N_PORT
A35	HDMI_TX0P_PORT		输出	HDMI_TX0P_PORT
A36	HDMI_TX1N_PORT		输出	HDMI_TX1N_PORT
A37	HDMI_TX1P_PORT		输出	HDMI_TX1P_PORT
A38	HDMI_TX2N_PORT		输出	HDMI_TX2N_PORT
A39	HDMI_TX2P_PORT		输出	HDMI_TX2P_PORT
A40	SPKN_OUT		输出	单声道喇叭输出负端 =0.6W
A41	SPKP_OUT		输出	单声道喇叭输出正端 =0.6W
A42	HPL_OUT		输出	耳机左声道输出
A43	HPL_SNS		输出	耳机接地
A44	HPR_OUT		输出	耳机右声道输出
A45	MIC1_INP		输入	麦克风正端输入
A46	MIC1_INN		输入	麦克风负端输入
A47	VDC		输入	上电自启动
A48	GND	GND		

B 端口如下:

PIN	GPIO 功能	IO 电平	IO 状态	IO 多功能复用定义
B1	VCC3V3_SYS		输入	主电源输入 3.3V/3A
B2	RESET_KEY	IO=3V3	输入	复位键 (备注:外部不能接上拉)
B3	PWRON_KEY	IO=3V3	输入	开关机键 (备注:外部不能接上拉)
B4	VCCIO_ACODEC	3V3	输出	电源输出 LDO4 POWER OUT default 3V3
B5	VCC_3V3	3V3	输出	电源输出 RK809 SW1 POWER OUT default 3V3
B6	VCC_1V8	1V8	输出	电源输出 RK809 BUCK5 POWER OUT default 1V8
B7	REFCLK_OUT_CAM	IO=3V3	输入输出	REFCLK_OUT/GPIO0_A0_d
B8	SDMMC0_DET_L	IO=3V3	输入输出	SDMMC0_DET/SATA_CP_DET/PCIE30X1_CLKREQn_M0/GPIO0_A4_u

Original engineering table, source page 18. Signal names are preserved; see the English legend on the preceding page.

Pin definition reference - sheet 6 of 11

B9	UART0_RX	IO=3V3	输入输出	PWM1_M0/GPUAVS/UART0_RX/GPIO0_C0_d
B10	UART0_TX	IO=3V3	输入输出	PWM2_M0/NPUAVS/UART0_TX/MCU_JTAG_TDI/GPIO0_C1_d
B11	EDP_HPDIN_M1	IO=3V3	输入输出	PWM3_IR/EDP_HPDIN_M1/PCIE30X1_WAKEn_M0/MCU_JTAG_TMS/GPIO0_C2_d
B12	LCD0_BL_PWM4	IO=3V3	输入输出	PWM4/VOP_PWM_M0/PCIE30X1_PERSTn_M0/MCU_JTAG_TRSTn/GPIO0_C3_d
B13	SPIO_MISO_M0	IO=3V3	输入输出	PWM6/SPIO_MISO_M0/PCIE30X2_WAKEn_M0/GPIO0_C5_d
B14	SPIO_CS0_M0	IO=3V3	输入输出	PWM7_IR/SPIO_CS0_M0/PCIE30X2_PERSTn_M0/GPIO0_C6_d
B15	UART2_RX_M0_DEBUG	IO=3V3	输入输出	UART2_RX_M0/GPIO0_D0_u
B16	UART2_TX_M0_DEBUG	IO=3V3	输入输出	UART2_TX_M0/GPIO0_D1_u
B17	I2C1_SCL_TP	IO=3V3	输入输出	I2C1_SCL/CAN0_TX_M0/PCIE30X1_BUTTONRSTn/MCU_JTAG_TDO/GPIO0_B3_u
B18	I2C1_SDA_TP	IO=3V3	输入输出	I2C1_SDA/CAN0_RX_M0/PCIE20_BUTTONRSTn/MCU_JTAG_TCK/GPIO0_B4_u
B19	SPIO_CLK_M0	IO=3V3	输入输出	I2C2_SCL_M0/SPIO_CLK_M0/PCIE20_WAKEn_M0/PWM1_M1/GPIO0_B5_u
B20	SPIO_MOSI_M0	IO=3V3	输入输出	I2C2_SDA_M0/SPIO_MOSI_M0/PCIE20_PERSTn_M0/PWM2_M1/GPIO0_B6_u
B21	SATA2_TXP		输出	PCIE20_TXP/SATA2_TXP/QSGMII_TXP_M1
B22	SATA2_TXN		输出	PCIE20_TXN/SATA2_TXN/QSGMII_TXN_M1
B23	SATA2_RXN		输入	PCIE20_RXN/SATA2_RXN/QSGMII_RXN_M1
B24	SATA2_RXP		输入	PCIE20_RXP/SATA2_RXP/QSGMII_RXP_M1
B25	PCIE20_REFCLK_P		输出	PCIE20_REFCLKP
B26	PCIE20_REFCLK_N		输出	PCIE20_REFCLKN
B27	GND/PMIC_32KOUT_WIFI		输出	GND/PMIC_32KOUT_WIFI（此脚默认是接地，如需时钟功能再打开）
B28	USB3_HOST1_SSTXP		输出	USB3_HOST1_SSTXP/SATA1_TXP/QSGMII_TXP_M0
B29	USB3_HOST1_SSTXN		输出	USB3_HOST1_SSTXN/SATA1_TXN/QSGMII_TXN_M078
B30	USB3_HOST1_SSRXP		输入	USB3_HOST1_SSRXP/SATA1_RXP/QSGMII_RXP_M0
B31	USB3_HOST1_SSRXN		输入	USB3_HOST1_SSRXN/SATA1_RXN/QSGMII_RXN_M0
B32	USB3_HOST1_DP		输入输出	USB3_HOST1_DP
B33	USB3_HOST1_DM		输入输出	USB3_HOST1_DM
B34	USB3_OTG0_SSTXP		输出	USB3_OTG0_SSTXP/SATA0_TXP
B35	USB3_OTG0_SSTXN		输出	USB3_OTG0_SSTXN/SATA0_TXN

Original engineering table, source page 19. Signal names are preserved; see the English legend on the preceding page.

Pin definition reference - sheet 7 of 11

	TXN			
B36	USB3_OTG0_SS RXP		输入	USB3_OTG0_SSRXP/SATA0_RXP
B37	USB3_OTG0_SS RXN		输入	USB3_OTG0_SSRXN/SATA0_RXN
B38	USB3_OTG0_DP		输入输出	USB3_OTG0_DP
B39	USB3_OTG0_D M		输入输出	USB3_OTG0_DM
B40	USB3_OTG0_ID		输入	USB3_OTG0_ID
B41	USB3_OTG0_VB USDET		输入	USB3_OTG0_VBUSDET
B42	GND	GND		
B43	EDP_TX_D1N		输出	EDP_TX_D1N
B44	EDP_TX_D1P		输出	EDP_TX_D1P
B45	EDP_TX_D0N		输出	EDP_TX_D0N
B46	EDP_TX_D0P		输出	EDP_TX_D0P
B47	EDP_TX_AUXP		输出	EDP_TX_AUXP
B48	EDP_TX_AUXN		输出	EDP_TX_AUXN

C 端口如下 (在底层):

PIN	GPIO 功能	IO 电平	IO 状态	IO 多功能复用定义
C1	SDMMC0_D1	IO=3V3	输入输出	SDMMC0_D1/UART2_RX_M1/UART6_RX_M1/PWM9_M1/GPIO1_D6_u
C2	SDMMC0_D0	IO=3V3	输入输出	SDMMC0_D0/UART2_TX_M1/UART6_TX_M1/PWM8_M1/GPIO1_D5_u
C3	SDMMC0_CLK	IO=3V3	输出	SDMMC0_CLK/TEST_CLKOUT/UART5_TX_M0/CAN0_RX_M1/GPIO2_A2_d
C4	VCC3V3_SD	3V3	输出	RK809 SW2 POWER OUT default 3V3
C5	SDMMC0_CMD	IO=3V3	输出	SDMMC0_CMD/PWM10_M1/UART5_RX_M0/CAN0_TX_M1/GPIO2_A1_u
C6	SDMMC0_D3	IO=3V3	输入输出	SDMMC0_D3/ARMJTAG_TMS/UART5_RTSn_M0/GPIO2_A0_u
C7	SDMMC0_D2	IO=3V3	输入输出	SDMMC0_D2/ARMJTAG_TCK/UART5_CTSn_M0/GPIO1_D7_u
C8	eMMC_KEY	3V3-5V	输入	升级键(需接高电平)
C9	ETH0_REFCLKO_25M	IO=1V8	输入输出	I2S2_MCLK_M0/ETH0_REFCLKO_25M/UART7_RTSn_M0/SPI2_CLK_M0/GPIO2_C1_d
C10	GMAC0_MCLKINOUT	IO=1V8	输入输出	I2S2_SCLK_TX_M0/GMAC0_MCLKINOUT/UART7_CTSn_M0/SPI2_MISO_M0/GPIO2_C2_d
C11	GMAC0_RXCLK	IO=1V8	输入输出	SDMMC1_D2/GMAC0_RXCLK/UART7_RX_M0/GPIO2_A5_u
C12	GMAC0_RXDV_CRS	IO=1V8	输入输出	I2S2_LRCK_RX_M0/GMAC0_RXDV_CRS/UART6_CTSn_M0/SPI1_CS0_M0/GPIO2_C0_d
C13	GMAC0_RXD0	IO=1V8	输入输出	GMAC0_RXD0/UART1_CTSn_M0/SPI1_MISO_M0/GPIO2_B6_u
C14	GMAC0_RXD1	IO=1V8	输入输出	I2S2_SCLK_RX_M0/GMAC0_RXD1/UART6_RTSn_M0/SPI1_MOSI_M0/GPIO2_B7_d
C15	GMAC0_RXD2	IO=1V8	输入输出	SDMMC1_D0/GMAC0_RXD2/UART6_RX_M0/GPIO2_A3_u

Original engineering table, source page 20. Signal names are preserved; see the English legend on the preceding page.

Pin definition reference - sheet 8 of 11

C16	GMAC0_RXD3	IO=1V8	输入输出	SDMMC1_D1/GMAC0_RXD3/UART6_TX_M0/GPIO2_A4_u
C17	GMAC0_TXCLK	IO=1V8	输入输出	SDMMC1_CLK/GMAC0_TXCLK/UART9_TX_M0/GPIO2_B0_d
C18	GMAC0_TXEN	IO=1V8	输入输出	GMAC0_TXEN/UART1_RTSn_M0/SPI1_CLK_M0/GPIO2_B5_u
C19	GMAC0_TXD0	IO=1V8	输入输出	GMAC0_TXD0/UART1_RX_M0/GPIO2_B3_u
C20	GMAC0_TXD1	IO=1V8	输入输出	GMAC0_TXD1/UART1_TX_M0/GPIO2_B4_u
C21	GMAC0_TXD2	IO=1V8	输入输出	SDMMC1_D3/GMAC0_TXD2/UART7_TX_M0/GPIO2_A6_u
C22	GMAC0_TXD3	IO=1V8	输入输出	SDMMC1_CMD/GMAC0_TXD3/UART9_RX_M0/GPIO2_A7_u
C23	GMAC0_MDIO	IO=1V8	输入输出	I2S2_SDO_M0/GMAC0_MDIO/UART9_CTSn_M0/SPI2_CS0_M0/GPIO2_C4_d
C24	GMAC0_MDC	IO=1V8	输入输出	I2S2_LRCK_TX_M0/GMAC0_MDC/UART9_RTSn_M0/SPI2_MOSI_M0/GPIO2_C3_d
C25	UART8_RTSn_M0	IO=1V8	输入输出	SDMMC1_PWREN/I2C4_SDA_M1/UART8_RTSn_M0/CAN2_RX_M1/GPIO2_B1_d
C26	UART8_CTSn_M0	IO=1V8	输入输出	SDMMC1_DET/I2C4_SCL_M1/UART8_CTSn_M0/CAN2_TX_M1/GPIO2_B2_u
C27	UART8_TX_M0	IO=1V8	输入输出	I2S2_SDI_M0/GMAC0_RXER/UART8_TX_M0/SPI2_CS1_M0/GPIO2_C5_d
C28	UART8_RX_M0	IO=1V8	输入输出	CLK32K_OUT1/UART8_RX_M0/SPI1_CS1_M0/GPIO2_C6_d
C29	GND	GND		
C30	SARADC_VIN4	IO=1V8	输入	ADC4_BATT_TC_L
C31	SARADC_VIN0	IO=1V8	输入	ADC0_升级键
C32	USB2_HOST2_DP		输入输出	USB2_HOST2_DP
C33	USB2_HOST2_DM		输入输出	USB2_HOST2_DM
C34	USB2_HOST3_DP		输入输出	USB2_HOST3_DP
C35	USB2_HOST3_DM		输入输出	USB2_HOST3_DM
C36	SDMMC2_D1_M0	IO=1V8	输入输出	CIF_D1/EBC_SDDO1/SDMMC2_D1_M0/I2S1_SCLK_TX_M1/VOP_BT656_D1_M1/GPIO3_C7_d
C37	SDMMC2_D0_M0	IO=1V8	输入输出	CIF_D0/EBC_SDDO0/SDMMC2_D0_M0/I2S1_MCLK_M1/VOP_BT656_D0_M1/GPIO3_C6_d
C38	SDMMC2_CLK_M0	IO=1V8	输出	CIF_D5/EBC_SDDO5/SDMMC2_CLK_M0/I2S1_SDI1_M1/VOP_BT656_D5_M1/GPIO3_D3_d
C39	SDMMC2_CMD_M0	IO=1V8	输出	CIF_D4/EBC_SDDO4/SDMMC2_CMD_M0/I2S1_SDI0_M1/VOP_BT656_D4_M1/GPIO3_D2_d
C40	SDMMC2_D3_M0	IO=1V8	输入输出	CIF_D3/EBC_SDDO3/SDMMC2_D3_M0/I2S1_SDO0_M1/VOP_BT656_D3_M1/GPIO3_D1_d
C41	SDMMC2_D2_M0	IO=1V8	输入输出	CIF_D2/EBC_SDDO2/SDMMC2_D2_M0/I2S1_LRCK_TX_M1/VOP_BT656_D2_M1/GPIO3_D0_d
C42	WIFI_WAKE_HOST	IO=1V8	输入输出	CIF_D6/EBC_SDDO6/SDMMC2_DET_M0/I2S1_SDI2_M1/VOP_BT656_D6_M1/GPIO3_D4_d
C43	WIFI_REG_ON	IO=1V8	输入输出	CIF_D7/EBC_SDDO7/SDMMC2_PWREN_M0/I2S1_SDI3_M1/VOP_BT656_D7_M1/GPIO3_D5_d
C44	CIF_CLKOUT	IO=1V8	输入输出	CIF_CLKOUT/EBC_GDCLK/PWM11_IR_M1/GPIO4_C0_d
C45	CAN2_TX_M0	IO=1V8	输入输出	I2C2_SCL_M1/EBC_SDSHR/CAN2_TX_M0/I2S1_SDO3_M1/G

Original engineering table, source page 21. Signal names are preserved; see the English legend on the preceding page.

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				PIO4_B5_d
C46	CAN2_RX_M0	IO=1V8	输入输出	I2C2_SDA_M1/EBC_GDSP/CAN2_RX_M0/ISP_FLASH_TRIGIN/VOP_BT656_CLK_M1/GPIO4_B4_d
C47	I2C4_SCL_M0	IO=1V8	输入输出	I2C4_SCL_M0/EBC_GDOE/ETH1_REFCLKO_25M_M1/SPI3_CLK_M0/I2S2_SDO_M1/GPIO4_B3_d
C48	I2C4_SDA_M0	IO=1V8	输入输出	I2C4_SDA_M0/EBC_VCOM/GMAC1_RXER_M1/SPI3_MOSI_M0/I2S2_SDI_M1/GPIO4_B2_d

D 端口如下:

PIN	GPIO 功能	IO 电平	IO 状态	IO 多功能复用定义
D1	GMAC1_MCLKINOUT_M1	IO=1V8	输入输出	CIF_CLKIN/EBC_SDCLK/GMAC1_MCLKINOUT_M1/UART1_CTSn_M1/I2S2_SCLK_RX_M1/GPIO4_C1_d
D2	GMAC1_RXCLK_M1	IO=1V8	输入输出	CIF_D13/EBC_SDDO13/GMAC1_RXCLK_M1/UART7_RX_M2/PDM_SDI3_M1/GPIO4_A3_d
D3	GMAC1_RXDV_CRS_M1	IO=1V8	输入输出	ISP_PRELIGHT_TRIG/EBC_SDCE3/GMAC1_RXDV_CRS_M1/I2S1_SDO2_M1/GPIO4_B1_d
D4	GMAC1_RXD0_M1	IO=1V8	输入输出	CAM_CLKOUT0/EBC_SDCE1/GMAC1_RXD0_M1/SPI3_CS1_M0/I2S1_LRCK_RX_M1/GPIO4_A7_d
D5	GMAC1_RXD1_M1	IO=1V8	输入输出	CAM_CLKOUT1/EBC_SDCE2/GMAC1_RXD1_M1/SPI3_MISO_M0/I2S1_SDO1_M1/GPIO4_B0_d
D6	GMAC1_RXD2_M1	IO=1V8	输入输出	CIF_D11/EBC_SDDO11/GMAC1_RXD2_M1/PDM_SDI1_M1/GPIO4_A1_d
D7	GMAC1_RXD3_M1	IO=1V8	输入输出	CIF_D12/EBC_SDDO12/GMAC1_RXD3_M1/UART7_TX_M2/PDM_SDI2_M1/GPIO4_A2_d
D8	GMAC1_TXCLK_M1	IO=1V8	输入输出	CIF_D10/EBC_SDDO10/GMAC1_TXCLK_M1/PDM_CLK1_M1/GPIO4_A0_d
D9	GMAC1_TXEN_M1	IO=1V8	输入输出	ISP_FLASHTRIGOUT/EBC_SDCE0/GMAC1_TXEN_M1/SPI3_CS0_M0/I2S1_SCLK_RX_M1/GPIO4_A6_d
D10	GMAC1_TXD0_M1	IO=1V8	输入输出	CIF_D14/EBC_SDDO14/GMAC1_TXD0_M1/UART9_TX_M2/I2S2_LRCK_TX_M1/GPIO4_A4_d
D11	GMAC1_TXD1_M1	IO=1V8	输入输出	CIF_D15/EBC_SDDO15/GMAC1_TXD1_M1/UART9_RX_M2/I2S2_LRCK_RX_M1/GPIO4_A5_d
D12	GMAC1_TXD2_M1	IO=1V8	输入输出	CIF_D8/EBC_SDDO8/GMAC1_TXD2_M1/UART1_TX_M1/PDM_CLK0_M1/GPIO3_D6_d
D13	GMAC1_TXD3_M1	IO=1V8	输入输出	CIF_D9/EBC_SDDO9/GMAC1_TXD3_M1/UART1_RX_M1/PDM_SDI0_M1/GPIO3_D7_d
D14	GMAC1_MDIO_M1	IO=1V8	输入输出	CIF_VSYNC/EBC_SDOE/GMAC1_MDIO_M1/I2S2_SCLK_TX_M1/GPIO4_B7_d
D15	GMAC1_MDC_M1	IO=1V8	输入输出	CIF_HREF/EBC_SDLE/GMAC1_MDC_M1/UART1_RTSn_M1/I2S2_MCLK_M1/GPIO4_B6_d160
D16	GND	GND		
D17	GMAC1_INT	IO=3V3	输入输出	LCDC_D0/VOP_BT656_D0_M0/SPI0_MISO_M1/PCIE20_CLKREQn_M1/I2S1_MCLK_M2/GPIO2_D0_d
D18	GMAC1_RSTn	IO=3V3	输入输出	LCDC_D1/VOP_BT656_D1_M0/SPI0_MOSI_M1/PCIE20_WAKEn_M1/I2S1_SCLK_TX_M2/GPIO2_D1_d
D19	GMAC0_INT	IO=3V3	输入输出	LCDC_D2/VOP_BT656_D2_M0/SPI0_CS0_M1/PCIE30X1_CLKREQn_M1/I2S1_LRCK_TX_M2/GPIO2_D2_d

Original engineering table, source page 22. Signal names are preserved; see the English legend on the preceding page.

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D20	GMAC0_RSTn	IO=3V3	输入输出	LCDC_D3/VOP_BT656_D3_M0/SPIO_CLK_M1/PCIE30X1_WA KEn_M1/I2S1_SDI0_M2/GPIO2_D3_d
D21	PCIE30X2_PRSN T_L	IO=3V3	输入输出	LCDC_D7/VOP_BT656_D7_M0/SPI2_MISO_M1/UART8_TX_ M1/I2S1_SDO0_M2/GPIO2_D7_d
D22	PCIE30X2_CLKR EQn_M1	IO=3V3	输入输出	LCDC_D4/VOP_BT656_D4_M0/SPI2_CS1_M1/PCIE30X2_CLK REQn_M1/I2S1_SDI1_M2/GPIO2_D4_d
D23	PCIE30X2_WAK En_M1	IO=3V3	输入输出	LCDC_D5/VOP_BT656_D5_M0/SPI2_CS0_M1/PCIE30X2_WA KEn_M1/I2S1_SDI2_M2/GPIO2_D5_d
D24	PCIE30X2_PERS Tn_M1	IO=3V3	输入输出	LCDC_D6/VOP_BT656_D6_M0/SPI2_MOSI_M1/PCIE30X2_PE RSTn_M1/I2S1_SDI3_M2/GPIO2_D6_d
D25	GPIO3_A0	IO=3V3	输入输出	LCDC_CLK/VOP_BT656_CLK_M0/SPI2_CLK_M1/UART8_RX_ M1/I2S1_SDO1_M2/GPIO3_A0_d
D26	SPI1_CS0_M1	IO=3V3	输入输出	LCDC_D8/VOP_BT1120_D0/SPI1_CS0_M1/PCIE30X1_PERSTn _M1/SDMMC2_D0_M1/GPIO3_A1_d
D27	GPIO3_A2	IO=3V3	输入输出	LCDC_D9/VOP_BT1120_D1/GMAC1_TXD2_M0/I2S3_MCLK_ M0/SDMMC2_D1_M1/GPIO3_A2_d
D28	I2S3_SCLK_M0	IO=3V3	输入输出	LCDC_D10/VOP_BT1120_D2/GMAC1_TXD3_M0/I2S3_SCLK_ M0/SDMMC2_D2_M1/GPIO3_A3_d
D29	I2S3_LRCK_M0	IO=3V3	输入输出	LCDC_D11/VOP_BT1120_D3/GMAC1_RXD2_M0/I2S3_LRCK_ M0/SDMMC2_D3_M1/GPIO3_A4_d
D30	I2S3_SDO_M0	IO=3V3	输入输出	LCDC_D12/VOP_BT1120_D4/GMAC1_RXD3_M0/I2S3_SDO_ M0/SDMMC2_CMD_M1/GPIO3_A5_d
D31	I2S3_SDI_M0	IO=3V3	输入输出	LCDC_D13/VOP_BT1120_CLK/GMAC1_TXCLK_M0/I2S3_SDI_ M0/SDMMC2_CLK_M1/GPIO3_A6_d
D32	PCIECLKIC_OE_ H	IO=3V3	输入输出	LCDC_D14/VOP_BT1120_D5/GMAC1_RXCLK_M0/SDMMC2_ DET_M1/GPIO3_A7_d
D33	ETH1_REFCLKO_ 25M_M0	IO=3V3	输入输出	LCDC_D15/VOP_BT1120_D6/ETH1_REFCLKO_25M_M0/SDM MC2_PWREN_M1/GPIO3_B0_d
D34	UART4_RX_M1	IO=3V3	输入输出	LCDC_D16/VOP_BT1120_D7/GMAC1_RXD0_M0/UART4_RX_ M1/PWM8_M0/GPIO3_B1_d
D35	UART4_TX_M1	IO=3V3	输入输出	LCDC_D17/VOP_BT1120_D8/GMAC1_RXD1_M0/UART4_TX_ M1/PWM9_M0/GPIO3_B2_d
D36	I2C5_SCL_M0	IO=3V3	输入输出	LCDC_D18/VOP_BT1120_D9/GMAC1_RXDV_CRS_M0/I2C5_S CL_M0/PDM_SDI0_M2/GPIO3_B3_d
D37	I2C5_SDA_M0	IO=3V3	输入输出	LCDC_D19/VOP_BT1120_D10/GMAC1_RXER_M0/I2C5_SDA_ M0/PDM_SDI1_M2/GPIO3_B4_d
D38	GPIO3_B5	IO=3V3	输入输出	LCDC_D20/VOP_BT1120_D11/GMAC1_TXD0_M0/I2C3_SCL_ M1/PWM10_M0/GPIO3_B5_d
D39	GPIO3_B6	IO=3V3	输入输出	LCDC_D21/VOP_BT1120_D12/GMAC1_TXD1_M0/I2C3_SDA_ M1/PWM11_IR_M0/GPIO3_B6_d
D40	UART3_TX_M1	IO=3V3	输入输出	LCDC_D22/PWM12_M0/GMAC1_TXEN_M0/UART3_TX_M1/ PDM_SDI2_M2/GPIO3_B7_d
D41	UART3_RX_M1	IO=3V3	输入输出	LCDC_D23/PWM13_M0/GMAC1_MCLKINOUT_M0/UART3_R X_M1/PDM_SDI3_M2/GPIO3_C0_d
D42	SPI1_MOSI_M1	IO=3V3	输入输出	LCDC_HSYNC/VOP_BT1120_D13/SPI1_MOSI_M1/PCIE20_PE RSTn_M1/I2S1_SDO2_M2/GPIO3_C1_d
D43	SPI1_MISO_M1	IO=3V3	输入输出	LCDC_VSYNC/VOP_BT1120_D14/SPI1_MISO_M1/UART5_TX

Original engineering table, source page 23. Signal names are preserved; see the English legend on the preceding page.

Pin definition reference - sheet 11 of 11

				_M1/I2S1_SDO3_M2/GPIO3_C2_d
D44	SPI1_CLK_M1	IO=3V3	输入输出	LCDC_DEN/VOP_BT1120_D15/SPI1_CLK_M1/UART5_RX_M1/I2S1_SCLK_RX_M2/GPIO3_C3_d
D45	UART7_TX_M1	IO=3V3	输入输出	PWM14_M0/VOP_PWM_M1/GMAC1_MDC_M0/UART7_TX_M1/PDM_CLK1_M2/GPIO3_C4_d
D46	UART7_RX_M1	IO=3V3	输入输出	PWM15_IR_M0/SPDIF_TX_M1/GMAC1_MDIO_M0/UART7_RX_M1/I2S1_LRCK_RX_M2/GPIO3_C5_d
D47	CAN1_RX_M1	IO=3V3	输入输出	PWM14_M1/SPI3_CLK_M1/CAN1_RX_M1/PCIE30X2_CLKRE_Qn_M2/I2S3_MCLK_M1/GPIO4_C2_d
D48	CAN1_TX_M1	IO=3V3	输入输出	PWM15_IR_M1/SPI3_MOSI_M1/CAN1_TX_M1/PCIE30X2_WAKEn_M2/I2S3_SCLK_M1/GPIO4_C3_d

E 端口如下:

PIN	GPIO 功能	IO 电平	IO 状态	IO 多功能复用定义
E1	PDM_SDI1_M0_ADC	IO=3.3V	输入输出	I2S1_SDO3_M0/I2S1_SDI1_M0/PDM_SDI1_M0/PCIE20_PERSTn_M2/GPIO1_B2_d
E2	GND	GND		
E3	SARADC_VIN2_HP_HOOK	IO=1.8V	输入	SARADC_VIN2
E4	PDM_SDI3_M0_ADC	IO=3.3V	输入输出	I2S1_SDO1_M0/I2S1_SDI3_M0/PDM_SDI3_M0/PCIE20_CLKREQn_M2/ACODEC_DAC_DATAR/GPIO1_B0_d
E5	PDM_SDI2_M0_ADC	IO=3.3V	输入输出	I2S1_SDO2_M0/I2S1_SDI2_M0/PDM_SDI2_M0/PCIE20_WAKEn_M2/ACODEC_ADC_SYNC/GPIO1_B1_d
E6	SARADC_VIN5	IO=1.8V	输入	SARADC_VIN5
E7	SARADC_VIN6	IO=1.8V	输入	SARADC_VIN6
E8	PDM_CLK1_M0_ADC	IO=3.3V	输入输出	I2S1_SCLK_RX_M0/UART4_RX_M0/PDM_CLK1_M0/SPDIF_TX_M0/GPIO1_A4_d
E9	I2C3_SDA_M0	IO=3.3V	输入输出	I2C3_SDA_M0/UART3_RX_M0/CAN1_RX_M0/AUDIOPWM_LOUT_P/ACODEC_ADC_DATA/GPIO1_A0_u
E10	I2C3_SCL_M0	IO=3.3V	输入输出	I2C3_SCL_M0/UART3_TX_M0/CAN1_TX_M0/AUDIOPWM_LOUT_N/ACODEC_ADC_CLK/GPIO1_A1_u
E11	SARADC_VIN7	IO=1.8V	输入	SARADC_VIN7
E12	BT_WAKE_HOST_H_GPIO0_B7	IO=1.8V	输入输出	PWM0_M0/CPUAVS/GPIO0_B7_d
E13	SDMMC0_PWREN_L	IO=3.3V	输入输出	SDMMC0_PWREN/SATA_MP_SWITCH/PCIE20_CLKREQn_M0/GPIO0_A5_d
E14	DVP_PWREN0_H_GPIO0_D3	IO=1.8V	输出	GPIO0_D3_d
E15	4G_PWREN_H_GPIO0_D5	IO=1.8V	输出	GPIO0_D5_d
E16	USB5V0_EN_H_GPIO0_D6	IO=1.8V	输出	GPIO0_D6_d
E17	HOST_WAKE_BT_H_GPIO0_B0	IO=1.8V	输入输出	CLK32K_IN/CLK32K_OUT0/PCIE30X2_BUTTONRSTn/GPIO0_B0_u

Original engineering table, source page 24. Signal names are preserved; see the English legend on the preceding page.

8. Document scope and next step

This is an English selection and integration reference. It intentionally excludes supplier branding, contact details, service claims and unverified commercial statements. Original photos and engineering-table facsimiles are included where they help identification or pin evaluation, with supplier marks redacted. This manual is not a replacement for the controlled pinout, schematic, mechanical drawing, processor datasheet or BSP documentation for a released production configuration.

Recommended next step

Share the product application, display size and interface, target resolution, operating system, required I/O, annual volume range, environmental conditions and development capability. The module can then be matched against the real project requirements and compared with chip-level or more integrated HMI paths.

Discuss a similar project

Visit openhmi.network to request relevant platform, display, touch and engineering expertise for your application.

Document control

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